

5 The lack of some inorganic ions in soil can affect the yield of fruit from crop plants.

- (a) The photograph shows part of a vine plant growing in soil lacking magnesium ions.

The vine plant shown has yellow leaves and reduced growth.



(Source: Nigel Cattlin / Alamy Stock Photo)

- (i) Which tissue is involved in the transport of magnesium ions in a plant?

(1)

- ☐ **A** amyloplast
- ☐ **B** plasmodesmata
- ☐ **C** sclerenchyma
- ☐ **D** xylem



- (ii) Explain how a lack of magnesium ions could result in yellow leaves and reduced growth.

(3)

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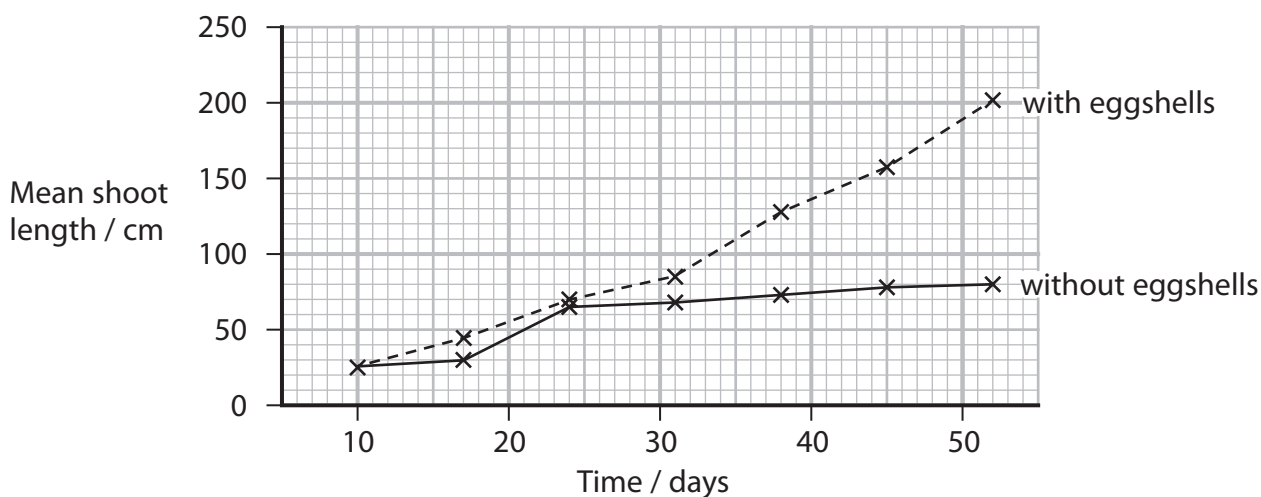
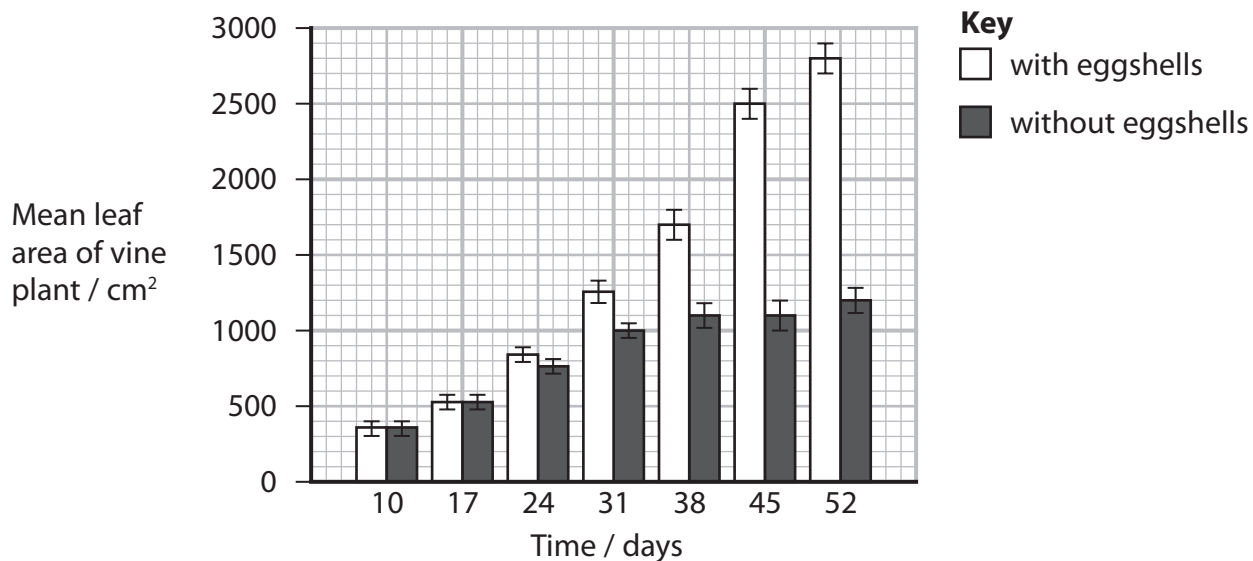
*(b) Calcium ions are also needed for the growth of vine plants and fruit production.

Chicken eggshells contain calcium ions. Some farmers add crushed eggshells to the soil surrounding vine plants.

The effect of adding crushed eggshells to the soil surrounding vine plants was investigated.

The mean leaf area, mean shoot length and mean chlorophyll content were calculated over 52 days.

The graphs show the results of this investigation.



(6)

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(Total for Question 5 = 10 marks)

